Scheduling 2: Study Notes

Course: Personal Training Certification

Topic: Scheduling, part 2 of 2 (analysing and designing a workout split)

About these notes: Short revision version of the Scheduling 2 lecture. For the full explanation of every concept, example and class question, see the Detailed Notes. Sources: the lecture recording, the course slides ("Scheduling"), the board photos and the published graphs shown in class.

Big idea: The number of main sets per muscle, the recovery needed, and how often the muscle is trained all fit together. Fewer sets work if you train the muscle more often, because each set adds less (diminishing returns) and the stimulus fades unless the muscle is trained again. Full body, upper/lower and PPL/bro splits are three frequency categories, and any training level can use any of them if the set range is respected.

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1. The master table

Main sets per muscle	Recovery period	Training frequency	Example splits	Ideal set range (upper cap)
1 to 3	48 h	Thrice a week (3x)	Full body (FBW) M: F W: F F: F Sun: rest	1 to 2 (max 3)
4 to 5	72 h	Twice a week (2x)	Upper / Lower M: U T: L Th: U F: L Sun: rest	2 to 3 (max 5)
More than 5	1 week	Once a week (1x)	Push-Pull-Legs (as shown once a week) Bro split (one body part a day)	6 to 8 (max 10)

Sets are MAIN sets per muscle per session. Frequency is counted per muscle, not per weight-training day.

Main sets per muscle	Recovery	Weekly outlook	Frequency	Splits	Ideal sets	Upper limit
1 to 3	48 h	Mon, Wed, Fri, rest Sun	3x a week	Full body (FBW)	1 to 2	3
4 to 5	72 h	Mon, Thu, rest Sun	2x a week	Upper/lower (U/L)	2 to 3	5
More than 5	1 week	Mon, rest Sun, next Mon	1x a week	PPL, bro split	6 to 8	10

- Sets are **main sets per muscle** (not warm-up sets, not all exercises together).
- **Frequency is per muscle, not per WT day.** Upper/lower has four WT days, but each muscle is trained twice.
- A non-WT day is not a rest day: it can be used for cardio. Sunday is the one mandatory rest day.
- A **workout split** is how your weight training sessions are arranged across the week, following some logic.

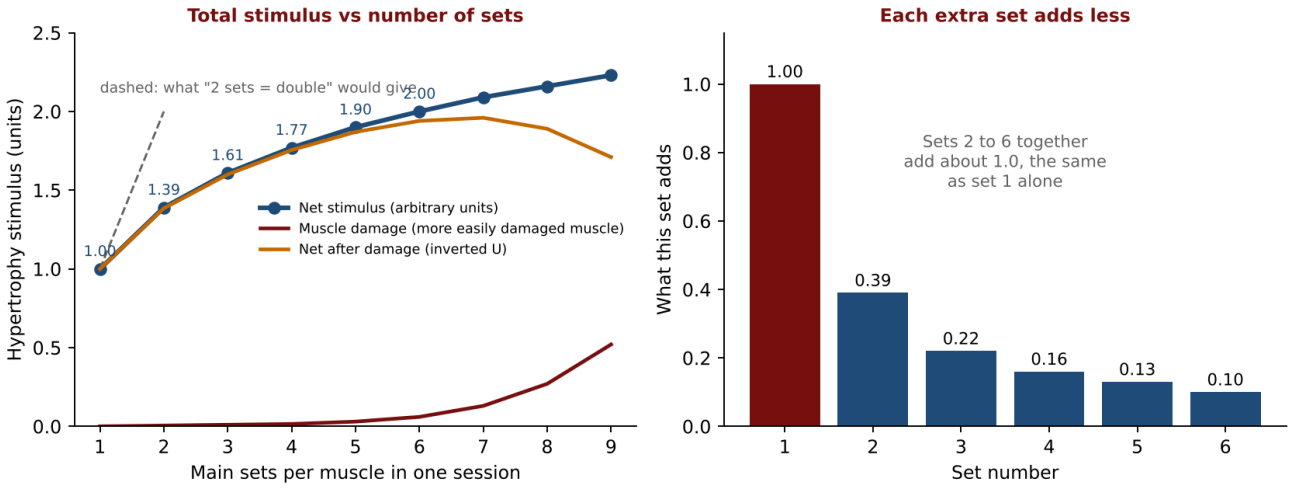
2. Types of split

Split	Logic	Example week
Full body	Whole body every session	Mon F, Wed F, Fri F, Sun R
Upper/lower	Upper body muscles on upper days, lower body on lower days	Mon U, Tue L, Thu U, Fri L, Sun R
Push, pull, legs	Grouped by movement pattern. Pull: back and biceps. Push: chest, shoulders, triceps. Legs	Mon P, Wed Pull, Fri L, Sat cardio, Sun R
Bro (body part) split	One body part per day; avoid shoulder work on back-to-back days	Back, chest, gap, legs, shoulders, arms, cardio, rest

3. Training level and F and I

- **F is proportional to 1/I:** when intensity is low, frequency can be high, and vice versa.
- Beginners train at low intensity (15 to 20 RM), so high frequency is fine.
- Beginners also need high frequency to build **neural pathways**, the **kinetic chain**, **proprioception**, grip and tightness, and the **Valsalva** breathing technique. Training a muscle once a week would stretch the 2 to 3 month beginner phase.
- Hence **full body, three days a week, is best for beginners**. The belief that full body is only for beginners is a myth: intermediate and advanced lifters can use it too, with the correct set range.

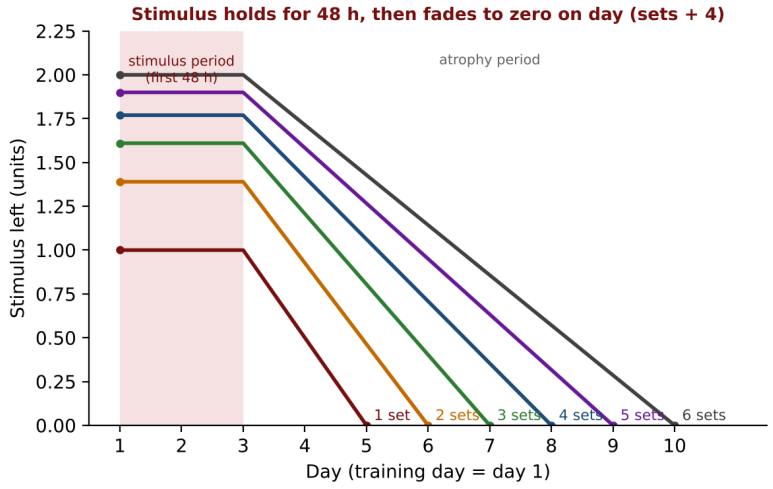
4. Factor 1: stimulus per set (diminishing returns)



Sets	Net stimulus	Added by this set
1	1.00	1.00
2	1.39	0.39
3	1.61	0.22
4	1.77	0.16
5	1.90	0.13
6	2.00	0.10

- The unit is arbitrary (AU). The lecture shows it as rupees to make it easier to relate to.
- "Added by this set" is the **difference between consecutive totals** (for example 1.61 minus 1.39 = 0.22).
- Six sets are needed to double the stimulus of one set. Sets 2 to 6 together add about 1.00, the same as set 1.
- Beyond six sets: 7 sets about 2.09, 8 about 2.16, 9 about 2.23.
- **Muscle damage** rises with more sets and pulls the net stimulus into an **inverted U**. This is why there are upper limits (max 3, 5, 10). More easily damaged muscles turn down earlier.

5. Factor 2: stimulus retention



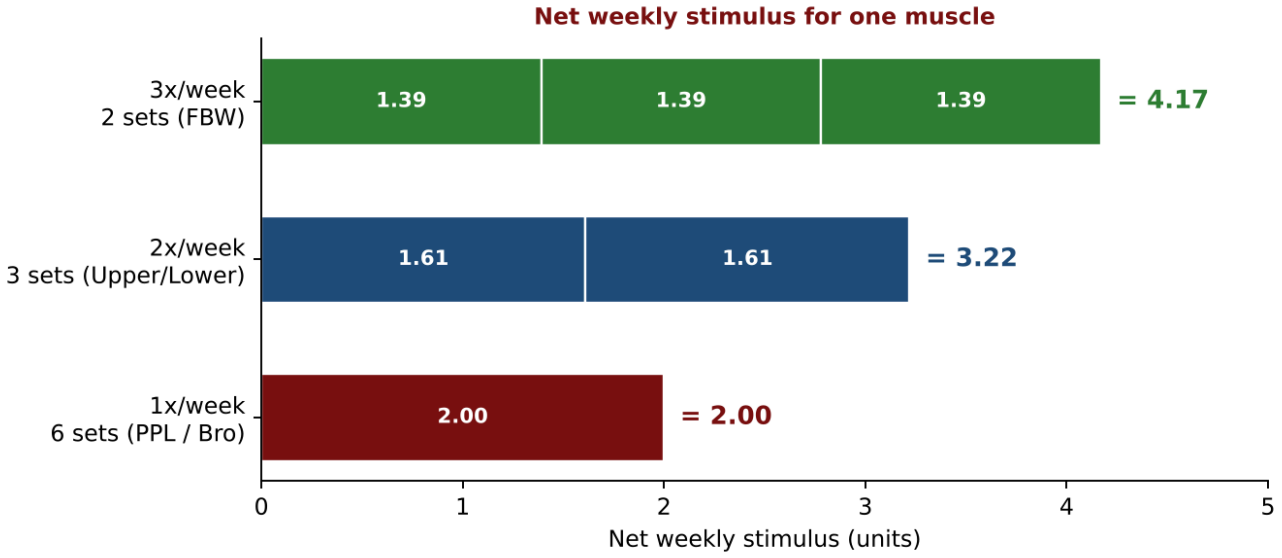
Sets	Net stimulus	Zero on	Last set adds day
1	1.00	Day 5	-
2	1.39	Day 6	+1 (0.39)
3	1.61	Day 7	+1 (0.22)
4	1.77	Day 8	+1 (0.16)
5	1.90	Day 9	+1 (0.13)
6	2.00	Day 10	+1 (0.10)

- Stimulus is not stored forever. If the muscle is not trained again, it reaches zero on **day (sets + 4)**, counting the training day as day 1.
- 1 set: day 5. 2 sets: day 6. 3 sets: day 7. 4 sets: day 8. 5 sets: day 9. 6 sets: day 10.
- Each extra set buys about **one more day**.
- This applies to every level; only the weight and the person differ.

6. Linking sets and frequency

- Low frequency with few sets means the stimulus is at zero before the next session, so nothing is carried forward.
- High frequency lets you keep the ball in the air with few sets per session.
- **Lower frequency, more sets. Higher frequency, fewer sets.**

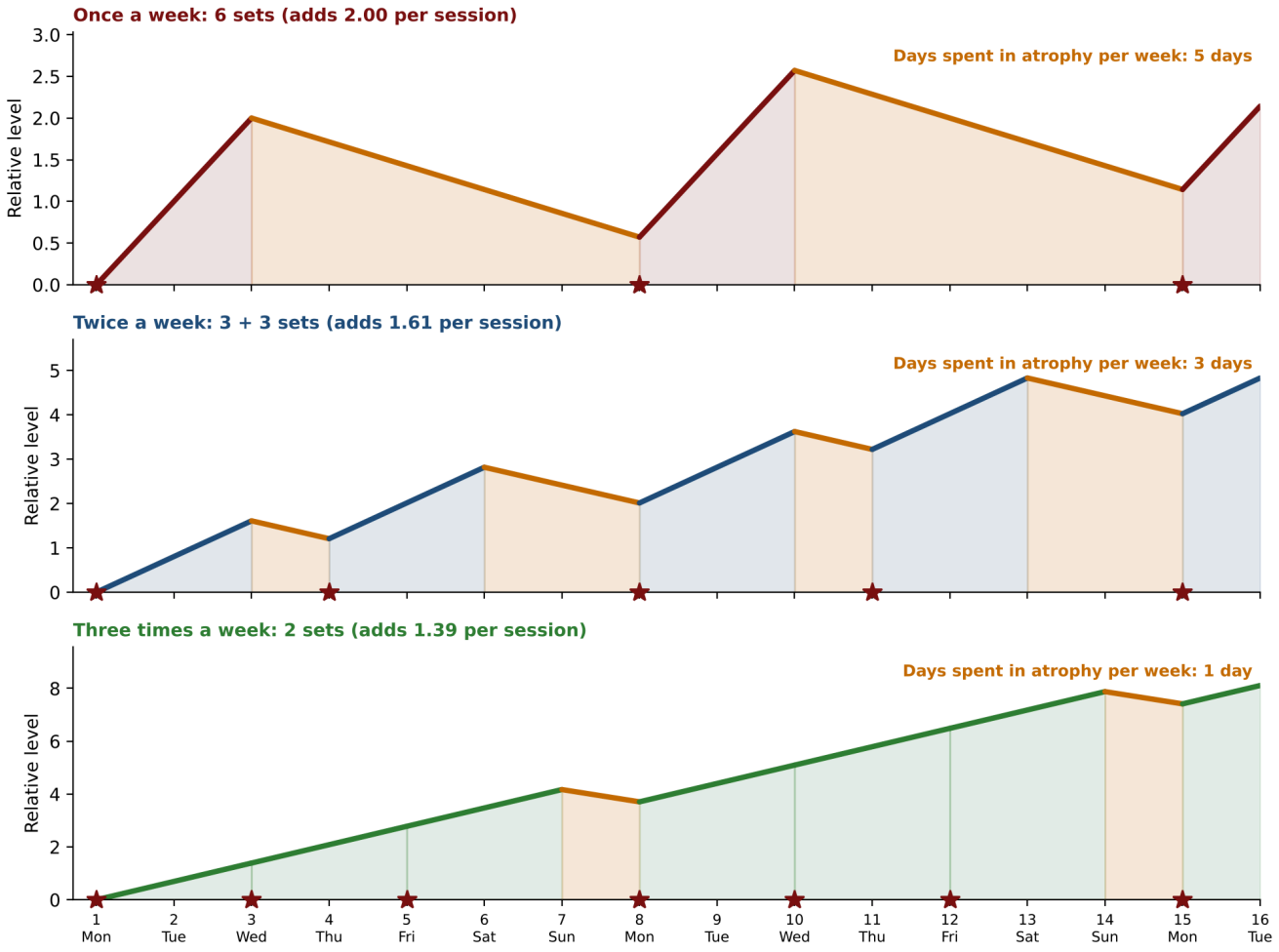
Split (frequency)	Sets per session	Net weekly stimulus
FBW (3x)	2	1.39 × 3 = **4.17**
Upper/lower (2x)	3	1.61 × 2 = **3.22**
PPL / bro (1x)	6	**2.00**



- Training a muscle three times a week gives the highest weekly stimulus.
- PPL and bro splits are not bad: they give room for many specific exercises per muscle, which suits bodybuilders.

7. Factor 3: determinants of frequency

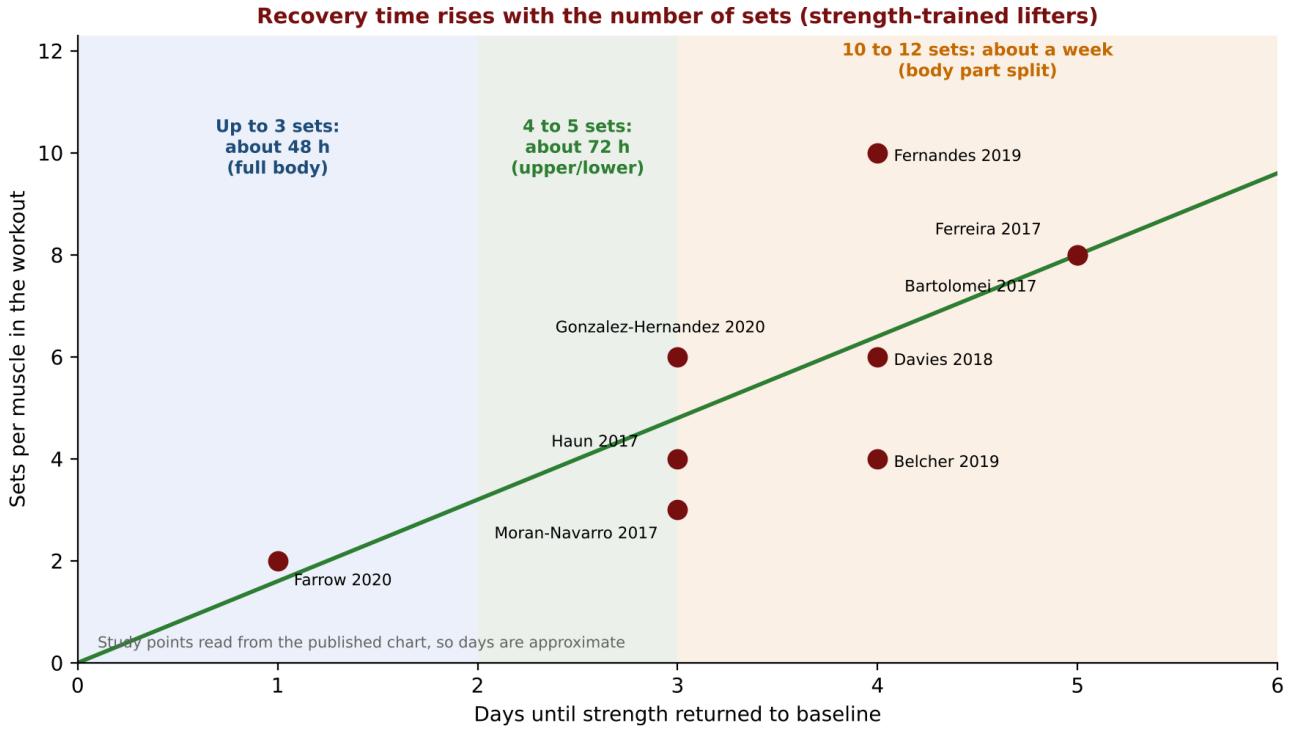
- **Hypertrophy** is growth of muscle cell size. **Atrophy** is shrinkage.
- **GAS** is general adaptation syndrome. **MyoPS** is myofibrillar protein synthesis (building the contractile proteins myosin and actin inside the myofibril).
- Hypertrophy (MyoPS) continues for the **first 48 hours** after training. After that, atrophy begins.
- Ball analogy: shoot the ball up again before it hits the ground.

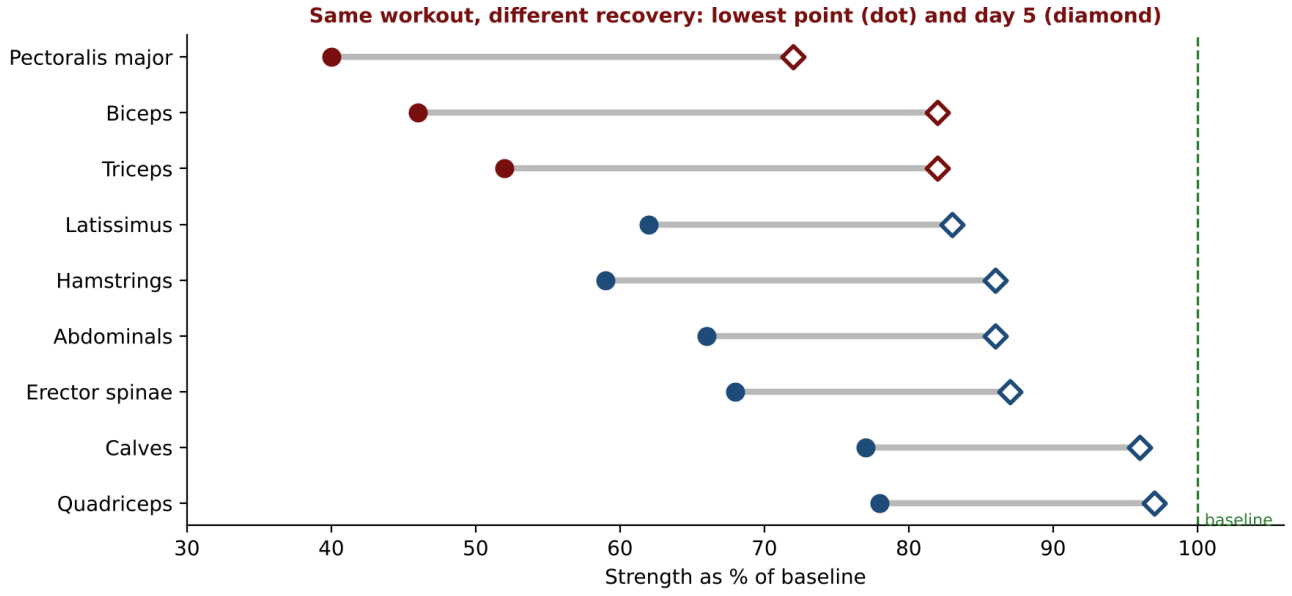


Coloured line = hypertrophy (rising for 48 h after each session). Orange line = atrophy (after the 48 h window, until the next session). Star = training day.
Each session starts from the balance left by the previous one. Schematic: only the shape matters, so each panel has its own vertical scale.

- Atrophy days per week in the figure: **5** (once a week), **3** (twice a week), **1** (three times a week).
- **FBW (3x):** the 48 hour window ends exactly on the next training day, so almost no atrophy occurs during the week.
- **U/L (2x):** the muscle is trained again on day 4, before zero (day 7).
- **PPL/bro (1x):** 6 sets last until day 10, and the next session (day 8) carries some balance forward.

The studies in one view





Red = more easily damaged muscles. Values are approximate, read from the published chart.

- Recovery time rises roughly in line with the number of sets: up to 3 sets about 48 h, 4 to 5 sets about 72 h, 10 to 12 sets about a week.
- Muscles differ: the pectoralis major, biceps and triceps recover slowest (more easily damaged). Quadriceps and calves recover fastest.
- Maintenance research: 3 sets once a week gives no gain and no loss; about 0.32 units are lost per day after the 48 h window.
- The figures are redrawn from the published charts (values approximate).

8. Sample programs (main sets only)

- Samples are not to be copied. Real programs are built around the client's goals, strengths, weaknesses, likes, time, preferred frequency and medical concerns. No program is "best" unless made for the specific client.
- Do 1 or 2 warm-up sets before the main sets.
- Count **synergists**: biceps work in rows and pulldowns, triceps work in presses. Frequency counts every exercise where the muscle works.
- A dedicated upper day gives room to hit all heads of the pectoralis major (flat for sternocostal and abdominal, incline for clavicular) and to add isolation.

Sample	Structure
Full body	Mon exercises alternate with Wed exercises (Fri repeats Mon, next Mon repeats Wed). Typical: 2 sets legs, 1 set calf, 3 sets row or pulldown, 3 sets chest press, 2 sets lateral raise or shoulder press, 2 sets biceps or triceps
Upper/lower	Mon upper, Tue lower, Wed off or cardio, Thu upper, Fri lower, Sat off or cardio, Sun off. Upper: 11 exercises. Lower: 8 exercises

9. Training levels in the slides

Level	Rep range	Split
Beginner	15 to 20 RM, no failure	Full body Mon, Wed, Fri; cardio + abs Tue, Thu, Sat
Intermediate	10 to 15 RM	Double split (upper/lower), alternating weekly
Advanced	6 to 8 RM	Triple split (push, pull, lower); four-day and six-day splits for higher levels

The slides also show linear periodisation (80% 5×5, 87% 4×4, 92% 3×3, 70% 3×10 de-load, then a new 1 RM).

10. Key terms

Term	Meaning
Workout split	How WT sessions are arranged across the week
Training frequency	How many times a muscle is trained per week
Main sets	Sets counted for a muscle, excluding warm-up sets
Stimulus per set (SPS)	How much one extra set adds to the stimulus
Law of diminishing returns	Each extra set adds less than the previous one
Stimulus retention	How many days a stimulus lasts if not trained again
Hypertrophy / atrophy	Increase / decrease in muscle cell size
MyoPS	Myofibrillar protein synthesis; raised for about 48 hours after training
GAS	General adaptation syndrome
Net weekly stimulus	Sum of the stimulus from each session in a week

11. Where the sources differ

1. **Reps.** The lecture mentions 4 to 6 RM stimulating reps; the slides give 10 to 15 RM (intermediate) and 6 to 8 RM (advanced).
2. **PPL.** The board shows PPL once a week (matching the slides' triple split). A six-day PPL trains each muscle twice, which by the lecture's rule would be the twice a week category (inference, not stated in the lecture).
3. **Sum of sets 2 to 6.** Said as "roughly 1.02"; the exact sum is 1.00.
4. **Study names and days.** Some were unclear in the recording; those in the detailed notes are read from the charts and are approximate.
5. **Lecture ended early.** The PPL and bro samples and the final level column were not delivered; the level table above comes from the slides.
6. **E-book samples** differ from the lecture's samples and rep ranges. Check with your course team.

12. Quick revision questions

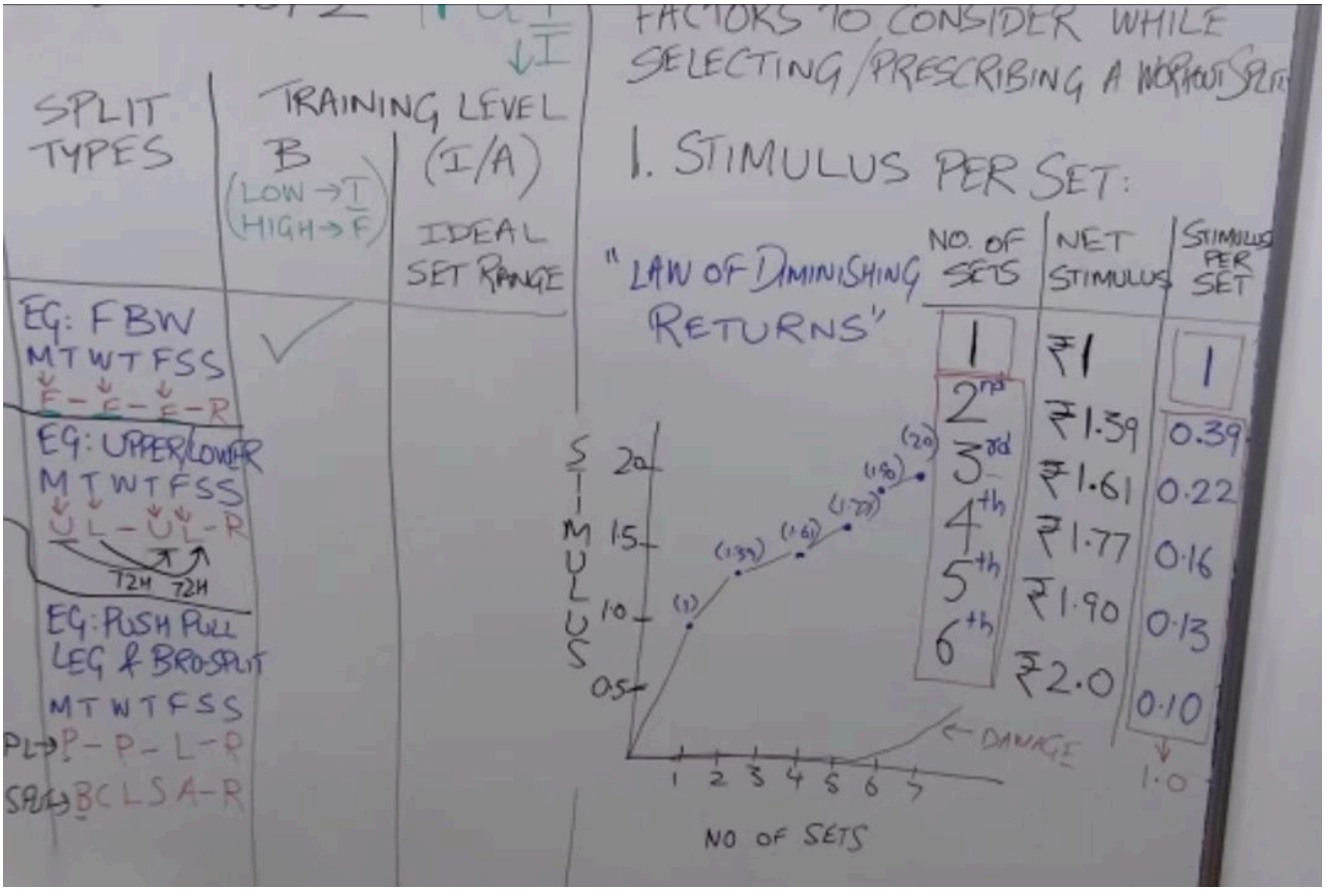
1. What does "No. of sets" mean in the master table?
2. Give the recovery period and training frequency for 1 to 3 sets, 4 to 5 sets, and more than 5 sets.
3. Why is frequency in an upper/lower split two and not four?
4. Which split fits each of the three scenarios?
5. How are muscles grouped in a push, pull, legs split?
6. What is a bro split, and why are shoulders kept away from back and chest days?
7. State the relationship between F and I.
8. Give three things a beginner must learn that need high frequency.

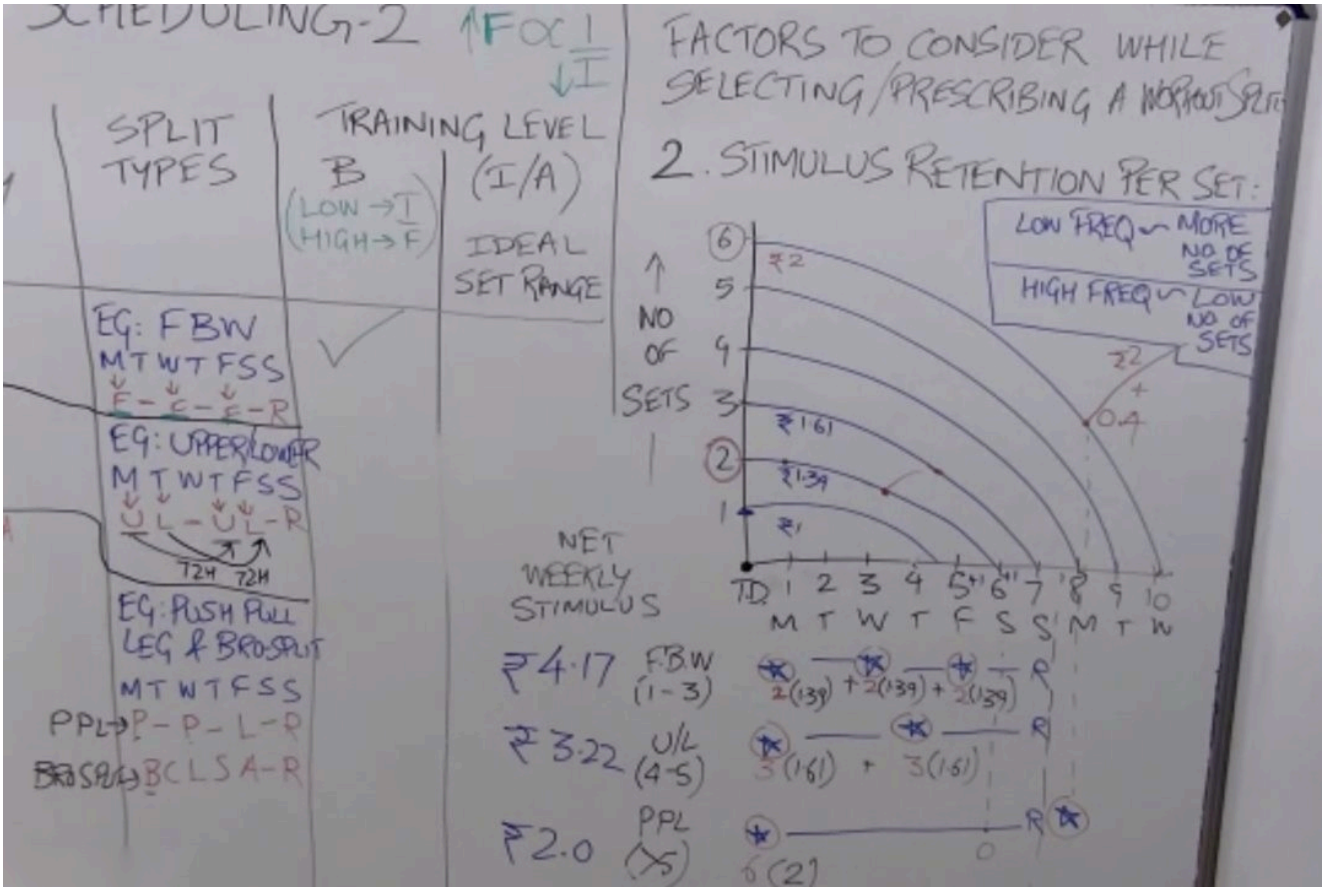
9. Why is full body the best choice for beginners, and why is "full body is only for beginners" a myth?
10. What stimulus does one set give, and how much do two sets give?
11. How is "stimulus per set" calculated for the third set?
12. How many sets are needed to double the stimulus of one set?
13. What is the law of diminishing returns?
14. Why does the net stimulus curve turn into an inverted U?
15. On which day does a stimulus reach zero if the muscle is not trained again?
16. Why can someone training thrice a week do fewer sets than someone training once a week?
17. Calculate the net weekly stimulus for 2 sets thrice a week, 3 sets twice a week, and 6 sets once a week.
18. What are GAS and MyoPS, and how long does MyoPS stay raised?
19. What is the "blessing in disguise" of the full body workout in terms of atrophy?
20. Give the ideal set range and upper limit for each frequency.

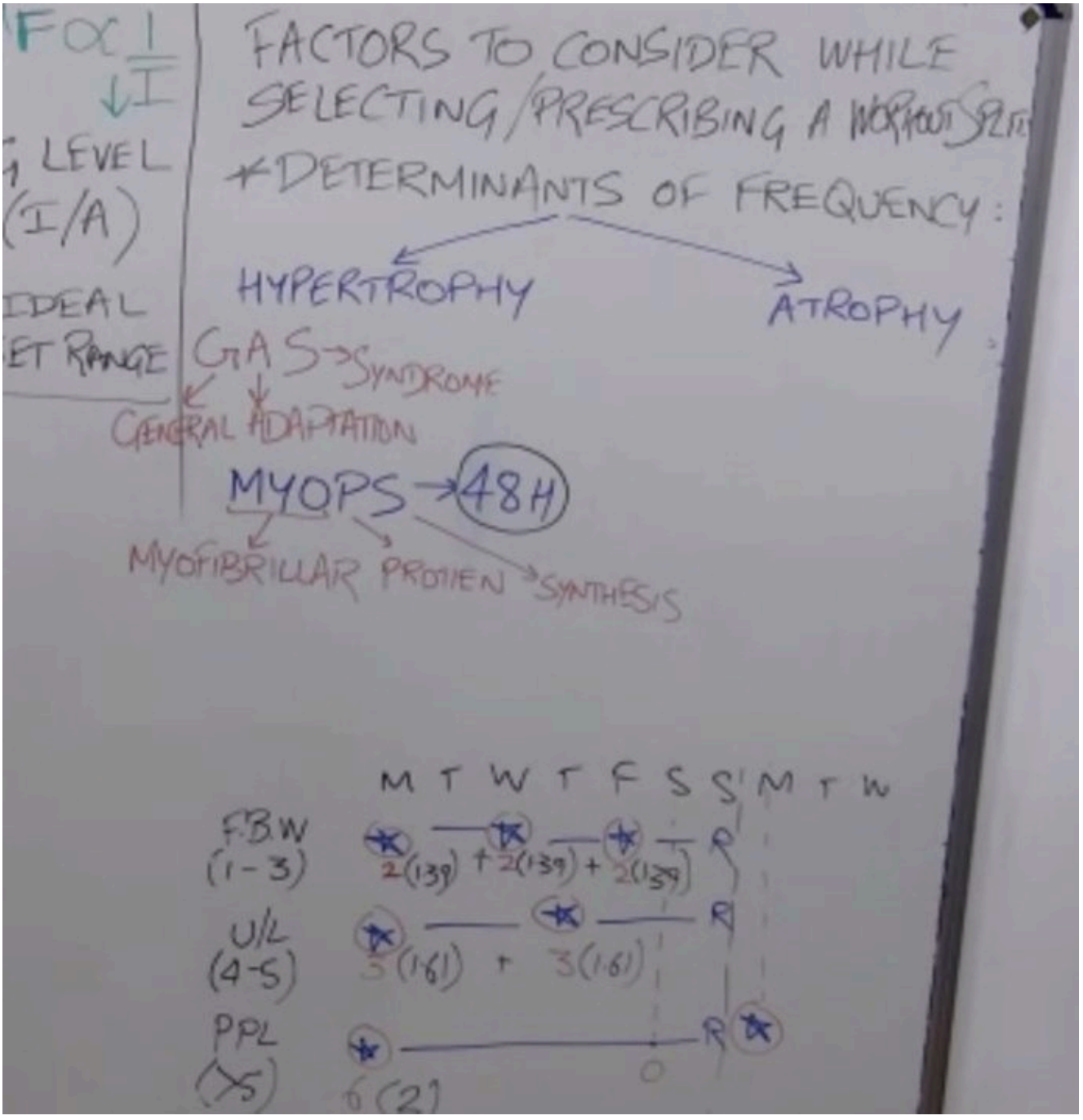
13. Answers

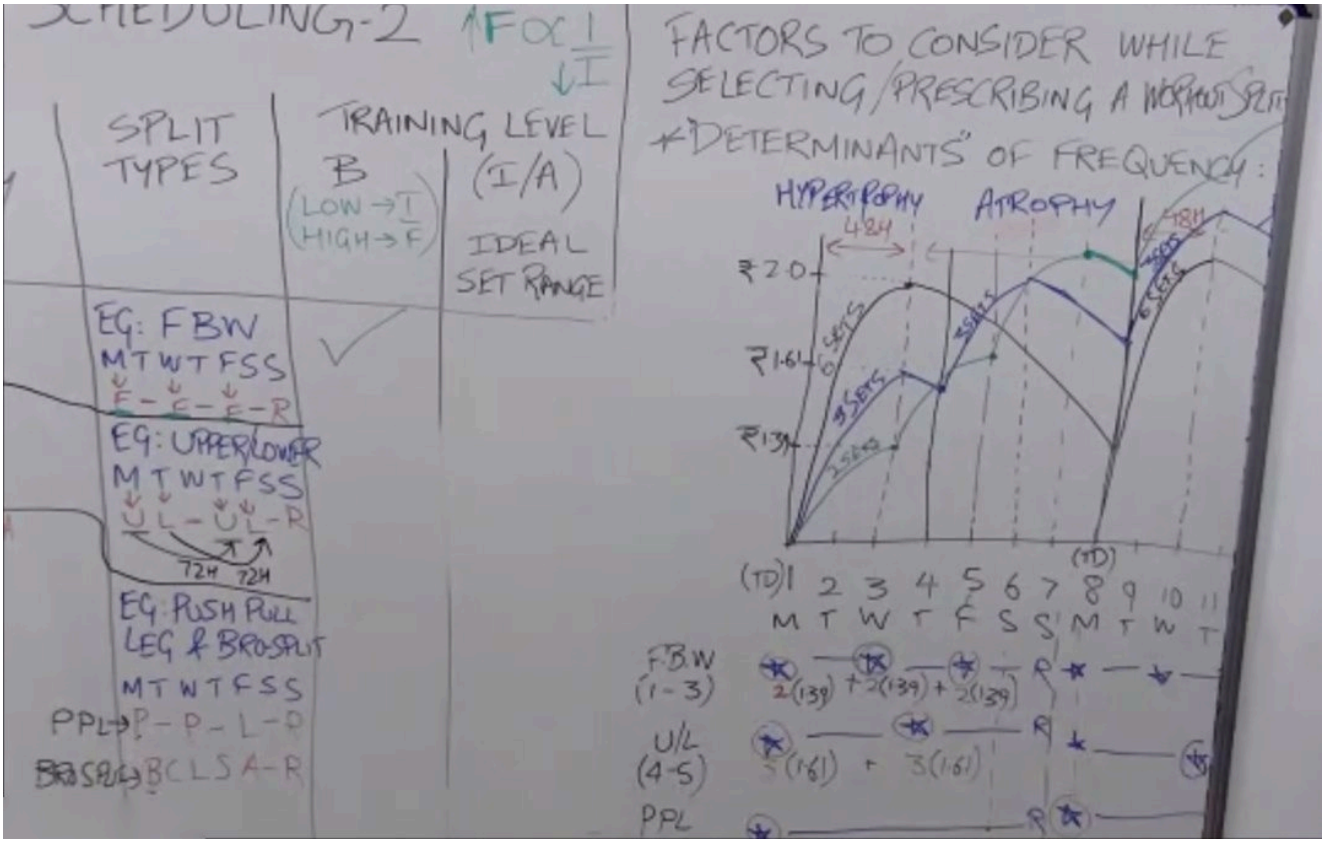
1. Main sets per muscle per session (not warm-up sets, not all exercises together).
2. 1 to 3 sets: 48 h, thrice a week. 4 to 5: 72 h, twice. More than 5: 1 week, once.
3. Frequency is counted per muscle. Each muscle is trained twice even though there are four WT days.
4. Full body, upper/lower, and PPL or bro split.
5. By movement pattern: pull (back, biceps), push (chest, shoulders, triceps), and legs.
6. One body part a day. Shoulders are kept apart so the joint is not used on back-to-back days.
7. F is proportional to 1/I: high frequency goes with low intensity.
8. Neural pathways, the kinetic chain, proprioception, grip and tightness, and the Valsalva technique (any three).
9. Low intensity allows high frequency, which speeds the learning phase. Intermediate and advanced lifters also gain with full body if they follow the set range.
10. 1.00 for one set; 1.39 for two.
11. 1.61 minus 1.39 = 0.22.
12. Six sets (2.00).
13. Each extra set adds less stimulus than the previous one, even though the total rises.
14. Muscle damage rises with sets and is deducted from the stimulus.
15. Day (sets + 4), with the training day as day 1.
16. Higher frequency retrains the muscle before the stimulus reaches zero, so fewer sets are enough; once a week must use more sets to carry some balance forward.
17. 4.17, 3.22 and 2.00.
18. General adaptation syndrome and myofibrillar protein synthesis; about 48 hours.
19. The day atrophy would begin is the day of the next session, so almost none occurs.
20. 3x a week: 1 to 2 (max 3). 2x: 2 to 3 (max 5). 1x: 6 to 8 (not above 10).

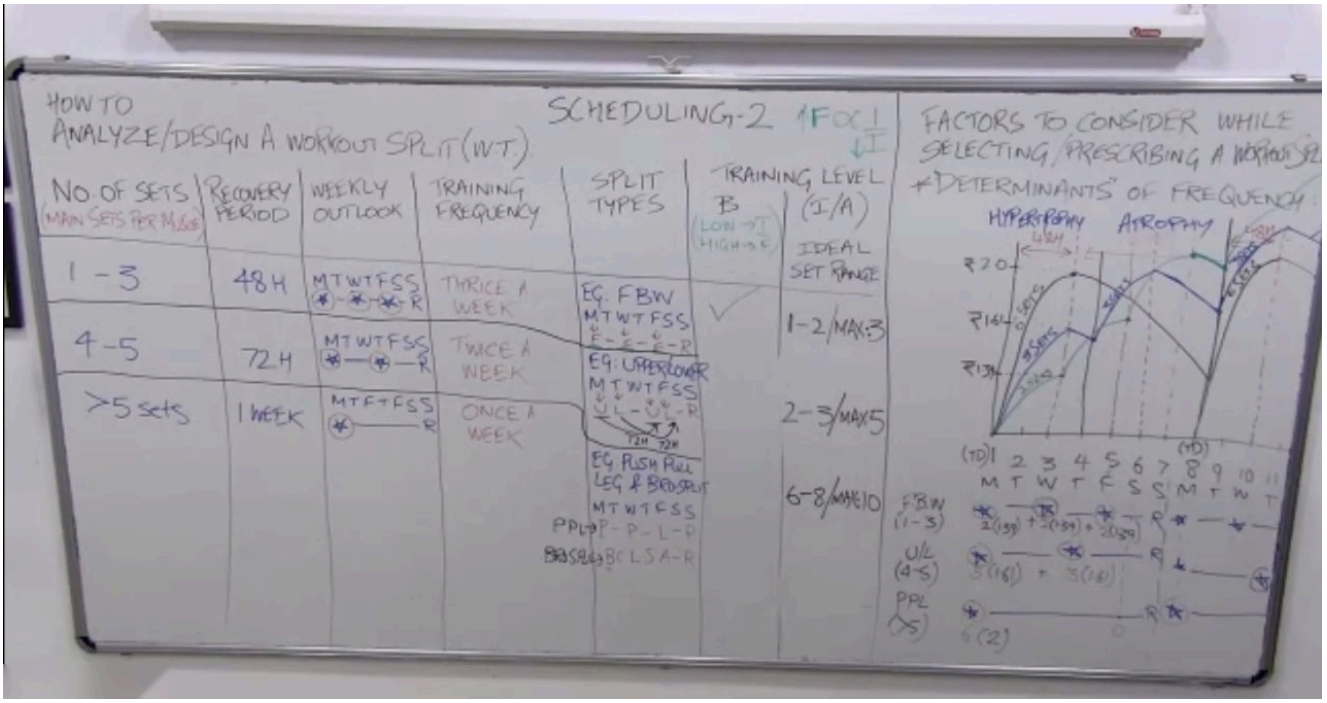
Board Photos from the Lecture






F ∝ 1/I
↓I
LEVEL
(I/A)
IDEAL
ET RANGE
FACTORS TO CONSIDER WHILE SELECTING/PRESCRIBING A WORKOUT SPLIT
*DETERMINANTS OF FREQUENCY:
HYPERTROPHY
ATROPHY
GAS → SYNDROME
GENERAL ADAPTATION
MYOPS → 48H
MYOFIBRILLAR PROTIEN SYNTHESIS
M T W T F S S M T W
F.B.W (1-3)
2(139) + 2(139) + 2(139)
U/L (4-5)
3(161) + 3(161)
PPL (X5)
6(2)


SCHEDULING-2
SPLIT TYPES
TRAINING LEVEL (I/A)
IDEAL SET RANGE
EG: FBW
EG: UPPER/LOWER
EG: PUSH PULL LEG & BRO SPLIT
FACTORS TO CONSIDER WHILE SELECTING/PRESCRIBING A WORKOUT SPLIT
DETERMINANTS OF FREQUENCY:
HYPERTROPHY
ATROPHY
48H
72H
FBW (1-3)
U/L (4-5)
PPL


HOW TO ANALYZE/DESIGN A WORKOUT SPLIT (WT.)
SCHEDULING-2
NO. OF SETS
RECOVERY PERIOD
WEEKLY OUTLOOK
TRAINING FREQUENCY
SPLIT TYPES
TRAINING LEVEL (I/A)
IDEAL SET RANGE
1-3
48H
THRICE A WEEK
4-5
72H
TWICE A WEEK
>5 sets
1 WEEK
ONCE A WEEK
1-2/MAX3
2-3/MAX5
6-8/MAX10
FACTORS TO CONSIDER WHILE SELECTING/PRESCRIBING A WORKOUT SPLIT
DETERMINANTS OF FREQUENCY:
HYPERTROPHY
ATROPHY